Assignment 2

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Part 1.

We split the data into training and test sets using a 60/40 ratio, as per the lecture slides. We then only kept the variables relevant for prediciton (the acoustic voice characteristics and the target variable) and excluded subject, age, sex, and test_time since they are not part of the modeling task. Then we scaled the data appropriately.

Part 2.

We computed a linear regression model using R's $lm()$ function on the scaled training data. We then calculated the estimated training and test Mean-Squared-Error (MSE) using the formula shown below.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Calculated values:

$\text{MSE}_{\text{train}}$ = **0.8785431**

MSE_{test} = **0.9354477**

From the model summary we can see which features are statistically significant. Features with a p-value $<$ 0.05 are typically considered significant.

Below is a table summarizing the estimated coefficients for each feature. The features are sorted by p-value in ascending order.

Table 1: Regression Coefficients

Feature	Estimate	Std. Error	t value	P value
DFA	-0.2803175	0.0201360	-13.9211999	< 1e-16
PPE	0.2264671	0.0328813	6.8874171	6.704e-12
HNR	-0.2385435	0.0363954	-6.5542224	6.411e-11
Shimmer.APQ11	0.3055462	0.0612360	4.9896520	6.343e-07
Jitter.Abs.	-0.1696093	0.0408055	-4.1565332	3.308e-05
NHR	-0.1853867	0.0455674	-4.0684029	4.838e-05
Shimmer.APQ5	-0.3875068	0.1137892	-3.4054787	0.0006679
Shimmer	0.5924358	0.2059810	2.8761675	0.0040496
Jitter...	0.1869308	0.1495605	1.2498674	0.2114314
Shimmer.dB.	-0.1726554	0.1393162	-1.2393066	0.2153149
Jitter.PPQ5	-0.0745680	0.0877659	-0.8496247	0.3955917
Shimmer.DDA	-32.3872408	77.1588144	-0.4197478	0.6746954
Shimmer.APQ3	32.0709324	77.1592423	0.4156460	0.6776945

Feature	Estimate	Std. Error	t value	P value
Jitter.RAP	-5.2695439	18.8341599	-0.2797865	0.7796578
Jitter.DDP	5.2495583	18.8375252	0.2786756	0.7805102
RPDE	0.0040681	0.0226637	0.1794988	0.8575564

Part 3.

We implemented the functions using these formulas:

Loglikelihood:

$$\log\, p(T|\theta,\sigma) = -\frac{N}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Ridge:

$$\text{Ridge}(\theta,\sigma,\lambda) = -\log\, p(T|\theta,\sigma) + \lambda\sum_{i=1}^{n}\theta_i^2$$

RidgeOpt:

$$\hat{\theta} = \arg\min_{\theta}\left\{\text{Ridge}(\theta,\sigma,\lambda)\right\}$$

DF:

$$\text{df}(\lambda) = \text{trace}\left(X\left(X^TX + \lambda I\right)^{-1}X^T\right)$$

Part 4.

For the last part, we compute the optimal θ paramaters for $\lambda \in \{1, 5, 100\}$.

Table 2: Performance Across Lambda Values

λ	$\text{MSE}_{\text{train}}$	MSE_{test}	DF
1	0.8786271	0.9349970	13.860736
100	0.8844104	0.9323317	9.924887
1000	0.9211079	0.9539555	5.643925

From Table 2 we can see the MSE scores and the degrees of freedom for each λ. The penalty paramater λ which is most appropriate is $\lambda = 100$. This penalty has the lowest test MSE (0.9323), and the training MSE is only slightly higher than $\lambda = 1$, so we are not over-penalizing. Its degrees of freedom (≈ 10) represent a model that is simpler than $\lambda = 1$ but not as restrictive as $\lambda = 1000$. The drop in DF at $\lambda = 1000$ leads to underfitting and worse test performance, and $\lambda = 1$, which has ≈ 4 more degrees of freedom, does not yield a better test MSE. This suggests that with $\lambda = 1$, the model is unnecessarily complex and more prone to overfitting.

ASSIGNMENT 4 QUESTION 3 I THINK

The cost function used in regression is typically the least squared error cost.

$$\frac{1}{n}\sum_{i=1}^{n}(\hat{y}(x_i;\theta)-y_i))^2$$

According to the book, this cost function can be expressed concisely in matrix form,

$$\frac{1}{n}\sum_{i=1}^{n}(\hat{y}(x_i;\theta)-y_i))^2=\frac{1}{n}\|\hat{y}-y\|_2^2=\frac{1}{n}\|X\theta-y\|_2^2=\frac{1}{n}\|\epsilon\|_2^2$$

where $\|\cdot\|$ denotes the usual Euclidean vector norm and $\|\cdot\|_2^2$ its square. [p.41]